

Python Foundation



Detailed
Course Syllabus

1. Python Basics

- a. Background Part 1 (IO Devices, CPU and Memory)
- b. Background Part 2 (Computer Organization)
- c. Why Do We Need Programming Languages
- d. Python Introduction
- e. Python Standard and Implementations
- f. How Python Programs Are Executed
- g. Python Programming Terminology
- h. Python Installation and First Program
- i. Comments in Python

2. Variables and Data Types

- a. Variables in Python
- b. Swap two Variables
- c. Id() in Python
- d. Type() in Python
- e. List Introduction
- f. Tuples in Python
- g. Set in Python
- h. Dictionary in Python
- i. Type Conversion in Python

3. Input and Output in Python

- a. print() in Python
- b. input() in Python

4. Operators

- a. Arithmetic Operators in Python
- b. Logical Operators in Python
- c. Identity Comparison Operators in Python
- d. Membership Test Operators in Python

- e. Bitwise Operators in Python Part 1
- f. Bitwise Operators in Python Part 2
- g. Arithmetic Progression nth Term in Python
- h. Geometric Progression nth Term in Python
- i. Sum of Natural numbers
- j. Find Last Digit in Python

5. Flow Control

- a. if, else and elif in Python
- b. Even-Odd in Python
- c. Largest of three in Python
- d. Leap Year in Python
- e. Calculator Program in Python

6. Functions

- a. Functions in Python
- b. Applications of Functions
- c. How Functions Work?
- d. Default Arguments
- e. Keyword Arguments
- f. Variable Length Arguments
- g. Parameter Passing in Python
- h. Returning Multiple Values in Python
- i. Global Variables in Python
- j. Find First Digit in Python
- k. Prime Factorization

7. Loops

- a. Loops In Python
- b. While Loops in Python
- c. range() in Python
- d. For Loop In Python
- e. Table of A Number
- f. Break In Python
- g. Continue In Python

- h. Nested Loop in Python
- i. Square Pattern in Python
- j. Printing Triangle Pattern in Python
- k. Inverted Triangle in Python
- l. Pyramid Pattern in Python
- m. Count Digits
- n. Factorial in Python
- o. GCD in Python
- p. LCM in Python
- q. Fibonacci Numbers in Python
- r. Check for Prime in Python
- s. All Divisors in Python
- t. Optimizations of All Divisors and Prime

8. String

- a. Strings in Python
- b. Escape Sequences and Raw Strings
- c. Formatted String in Python
- d. String Operations Part (1)
- e. String Operations Part (2)
- f. String Comparison in Python
- g. Pattern Searching in Python
- h. Check For Palindrome In Python
- i. Reverse A String in Python
- j. Decimal to Binary Conversion
- k. Binary to Decimal Conversion

9. List

- a. Slicing (List, Tuple And String)
- b. Get Smaller Elements
- c. Separate Even and Odd
- d. Comprehensions in Python
- e. Average or Mean of A List
- f. Count Distinct Elements in A List
- g. Check if a list is Sorted

10. Object-Oriented Programming (OOP)

- a. Introduction to OOPs
- b. Classes and Objects
- c. Encapsulation
- d. Class Instance Attributes
- e. Class Members Access
- f. Decorators
- g. Class Methods and Static Members
- h. Inheritance
- i. Types of Inheritance
- j. Multiple Inheritance
- k. Polymorphism
- l. Abstraction
- m. More Examples of Polymorphism
- n. Operator Overloading
- o. Abstract Class